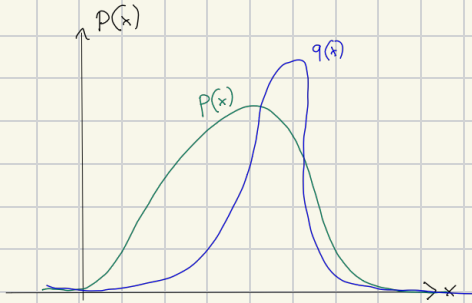




KL Divergence

- we know how to take distances on the
 - o Real Line
 - o Real Plane (conclusion 2D)



KL helps us find how similar p and q are

$$KL(p||q) = \mathbb{E}_{x \sim p(x)} \left[\log \left(\frac{p(x)}{q(x)} \right) \right] = \begin{cases} \sum_x p(x) \cdot \log \left(\frac{p(x)}{q(x)} \right) & \text{if } X \text{ is discrete} \\ \int_x p(x) \log \left(\frac{p(x)}{q(x)} \right) & \text{if } X \text{ is cont.} \end{cases}$$

Example

Weather (w) $\left\{ \begin{array}{l} \text{bad} \\ \text{good} \end{array} \right\}$ is bernoulli distributed $w \sim \text{Bernoulli}(\theta)$

probability of good weather $p(y=1|x)$, bad weather $p(y=0|x) = 1 - p(y=1|x)$

$W \in \{\text{Bad, Good}\}$

- now we have two opinions on W (two distributions)

Person 1
 $\theta_A = 0.8$
 $w \sim \text{Bernoulli}(\theta_A) = p(w)$

Person 2
 $\theta_B = 0.7$
 $w \sim \text{Bernoulli}(\theta_B) = q(w)$

$$D_{KL}(p||q) = \sum_{w \in \Omega} p(w) \log \left(\frac{p(w)}{q(w)} \right) = \sum_{w=0}^1 P(W=w) \log \left(\frac{p(W=w)}{q(W=w)} \right)$$

$$\left[\text{Bernoulli}(W; \theta) = \theta^w (1-\theta)^{1-w} \right]$$

$$= \theta_A^0 (1-\theta_A)^{1-0} \cdot \log \left(\frac{\theta_A^0 \cdot (1-\theta_A)^{1-0}}{\theta_B^0 \cdot (1-\theta_B)^{1-0}} \right) + \theta_A^1 \cdot (1-\theta_A)^{1-1} \cdot \log \left(\frac{\theta_A^1 \cdot (1-\theta_A)^{1-1}}{\theta_B^1 \cdot (1-\theta_B)^{1-1}} \right)$$

$$= 1 \cdot (1-\theta_A) \cdot \log \left(\frac{1 \cdot (1-\theta_A)^{1-0}}{1 \cdot (1-\theta_B)^{1-0}} \right) + \theta_A^1 \cdot 1 \cdot \log \left(\frac{\theta_A^1 \cdot 1}{\theta_B^1 \cdot 1} \right)$$

$$= (1-\theta_A) \cdot \log \left(\frac{(1-\theta_A)}{(1-\theta_B)} \right) + \theta_A \cdot \log \left(\frac{\theta_A}{\theta_B} \right)$$

$$= (1-0.8) \cdot \log \left(\frac{(1-0.8)}{(1-0.7)} \right) + 0.8 \cdot \log \left(\frac{0.8}{0.7} \right)$$

$$= \underline{\underline{0.257}}$$